# HUBERT MUCHALSKI

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## Appointments

**Professor of Chemistry** 2026–present
Department of Chemistry and Biochemistry
California State University, Fresno, CA

**Associate Professor of Chemistry** 2021–2026
Department of Chemistry and Biochemistry
California State University, Fresno, CA

**Visiting Associate Researcher** 2023
Department of Chemistry and Biochemistry
University of California Santa Barbara
*Host: Prof. Bruce H. Lipshutz*

**Assistant Professor of Chemistry** 2015–2021
Department of Chemistry and Biochemistry
California State University, Fresno, CA

**Visiting Scholar** 2015-2021
Department of Chemistry
Vanderbilt University, Nashville, TN

**Postdoctoral Scholar** 2012-2015
Department of Chemistry
Vanderbilt University, Nashville, TN
*Advisor: Prof. Ned A. Porter*

## Education

**Vanderbilt University,** Nashville, TN 2006–2012
Ph.D., Chemistry (Advisor: Prof. Jeffrey N. Johnston)

**Wroclaw University of Technology,** Wrocław, Poland 2001–2006
Magister, Chemistry (Advisor: Prof. Mirosław Giurg)

## Honors and Awards

**Outstanding Teaching Award** AY 2019
College of Science and Mathematics, Fresno State

PUBLICATIONS

**Refereed Journal Articles** (†undergraduate, ‡MS student)

1. Pisor, J.W.‡; Garcia, I. C.†; Mamo, K.†; **Muchalski, H.** Synthesis of benzofurans from THP acetals of 2-alkynylphenols catalyzed by gold(I)-NHC complexes. *Chem. Asian J.* **2025**, *92*, 153620.
http://dx.doi.org/10.1002/asia.202500093
2. Iler, K. S.; Yirak, J. R.†; **Muchalski, H.**; Lipshutz, B. H. Synthesis of S,S-Di(pyridin-2-yl)carbonodithioate (DPDTC) for the Reduction of Carboxylic Acids. *Org. Synth.* **2024**, *101*, 274–294.
http://www.orgsyn.org/demo.aspx?prep=v101p0274
3. Closser, K. D.; Hawker, M. J.; **Muchalski, H.** Quantized Grading: An Ab Initio Approach to Using Specifications Grading in Physical Chemistry. *J. Chem. Educ.* **2024**, *101* (2), 474–482.
https://doi.org/10.1021/acs.jchemed.3c00872
4. Le, Q.‡; Dillon, C. C.‡; Lichtenstein, D. A.†; Pisor, J.†; Closser, K. D. **Muchalski, H.** Gold(I)–NHC-catalysed synthesis of benzofurans via migratory cyclization of 2-alkynylaryl benzyl ethers *Org. Biomol. Chem.* **2020**, *28*, 8186–8191
http://dx.doi.org/10.1039/d0ob01538e
5. Rajaram, P.‡; Rivera, A. M.‡; Muthima, K.‡ Olveda, N.‡; **Muchalski, H.**; and Chen, Q.-C. Second-Generation Androgen Receptor Antagonists as Hormonal Therapeutics for Three Forms of Prostate Cancer *Molecules* **2020**, *20*, 2448.
https://doi.org/10.3390/molecules25102448
6. Dillon, C. C.†; Keophimphone, B.†; Sanchez, M.†; Kaur, P†.; **Muchalski, H.** Synthesis of 2-substituted benzo[b]thiophenes via gold(I)–NHC- catalyzed cyclization of 2-alkynyl thioanisoles *Org. Biomol. Chem.* **2018**, *16*, 9279–9284.
https://doi.org/10.1039/C8OB02196A
**Award**: Selected as Department's *Outstanding Publication* for 2018–2019 AY
7. Lamberson, C. R.; **Muchalski, H.**; McDuffee, K. B.†; Tallman, K. A.; Xu, L.; Porter, N. A.; Propagation rate constants for the peroxidation of sterols on the biosynthetic pathway to cholesterol *Chem. Phys, Lipids* **2017**, *207*, Part B, 51–58.
http://dx.doi.org/10.1016/j.chemphyslip.2017.01.006
8. **Muchalski, H.**; Site-Specific Synthesis and Application of Deuterium-Labeled Sterols. *ARKIVOC* **2017** part ii, 507–533.
https://doi.org/10.24820/ark.5550190.p009.755
9. **Muchalski, H.**.; Levonyak, A. J.†; Xu, L.; Ingold, K. U.; Porter, N. A. Competition H(D) Kinetic Isotope Effects in the Autoxidation of Hydrocarbons. *J. Am. Chem. Soc.* **2015**, *137*, 94–97.
http://dx.doi.org/10.1021/ja511434j
10. **Muchalski, H.**.; Xu, L.; Porter, N. A. Tunneling in Tocopherol-Mediated Peroxidation of 7-Dehydrocholesterol. *Org. Biomol. Chem.* **2015**, *13*, 1249–1253.
http://dx.doi.org/10.1039/C4OB02377C
11. Lamberson, C. R.; Xu, L.; **Muchalski, H.**.; Montenegro-Burke, J.R.; Shmanai, V. V.; Bekish, A. V.; McLean, J. A.; Clarke, C. F.; Shchepinov, M. S.; Porter, N. A. Unusual Kinetic Isotope Effects of Deuterium Reinforced Polyunsaturated Fatty Acids in Tocopherol-Mediated Free Radical Chain Oxidations. *J. Am. Chem. Soc.* **2014**,

*136*, 838–841.
http://dx.doi.org/10.1021/ja410569g

12. Giurg, M.; **Muchalski, H.**.; Kowal E. A. Oxofunctionalized *trans*-2-Carboxy-cinnamic Acids by Catalytic Domino Oxidation of Naphthols and Hydronaphthoquinones. *Synth. Commun.* **2012**, *42*, 2526–2539.
http://dx.doi.org/10.1080/00397911.2011.561945
13. Troyer, T. L.; **Muchalski, H.**.; Hong, K. B.; Johnston, J. N. Origins of Selectivity in Brønsted Acid Promoted Diazoalkane–Azomethine Reactions (The aza-Darzens Aziridine Synthesis). *Org. Lett.* **2011**, *13*, 1790–1792.
http://dx.doi.org/10.1021/ol200313m
14. **Muchalski, H.**.; Hong, K. B.; Johnston, J. N. Brønsted acid-promoted azide-olefin [3 + 2] cycloadditions for the preparation of contiguous aminopolyols: the importance of disiloxane ring size to a diastereoselective, bidirectional approach to zwittermicin A. *Beilstein J. Org. Chem.* **2010**, *6*, 1206–1210.
http://dx.doi.org/10.3762/bjoc.6.138
15. **Muchalski, H.**.; Troyer, T. L.; Doody, A. B.; Johnston, J. N. Preparation of isopropyl 2-diazoacetyl-(phenyl)carbamate. *Org. Synth.* **2011**, *Vol. 88*, 212–223.
16. Johnston, J. N.; Muchalski, H.; Troyer, T. L. Protonate or Alkylate: Stereoselective Brønsted Acid Catalysis of C–C Bond Formation Using Diazoalkanes. *Angew. Chem. Int. Ed.* **2010**, *49*, 2290–2298.
17. Troyer, T. L.; **Muchalski, H.**.; Johnston, J. N. Brønsted acid activation of α-diazo imide: a *syn*-glycolate Mannich reaction. *Chem. Commun.* **2009**, *32*, 6195–6197.
18. Adkins, C. T.; **Muchalski, H.**.; Harth, E. Nanoparticles with Individual Site-Isolated Semiconducting Polymers from Intramolecular Chain Collapse Processes. *Macromolecules* **2009**, *42*, 5786–5792.
19. Giurg, M.; Kowal, E. A.; **Muchalski, H.**.; Syper, L.; Młochowski, J. Catalytic oxidative domino degradation of alkyl phenols towards 2- and 3-substituted muconolactones. *Synth. Commun.* **2008**, *39*, 251–266.
20. Daniels, R. N.; Kim, K.; Lebois, E. P.; **Muchalski, H.**; Hughes, M.; Lindsley, C. W. Microwave-assisted protocols for the expedited synthesis of pyrazolo[1,5-a] and [3,4-d]pyrimi-dines. *Tetrahedron Lett.* **2008**, *49*, 305–310.
21. Niswender C. M.; Lebois E. P.; Luo Q.; Kim K.; **Muchalski, H.**; Yin H.; Conn P. J.; Lindsley C. W. Positive allosteric modulators of the metabotropic glutamate receptor subtype 4 (mGluR4): Part I. Discovery of pyrazolo[3,4-d]pyrimidines as novel mGluR4 positive allosteric modulators. *Bioorg. Med. Chem. Lett.* **2008**, *18*, 5626.
22. Croce, T. A.; Hamilton, S. K.; Chen, M. L.; **Muchalski, H.**.; Harth, E. M. Alternative *o*-Quinodimethane Cross-Linking Precursors for Intramolecular Chain Collapse Nano-particles. *Macromolecules* **2007**, *40*, 6028–6031.

**Book and Book Chapters**

1. Lipshutz, B.H.; **Muchalski, H.** *Organic Laboratory Experiments of the Future: Doing Chemistry in Water*; Elsevier Academic Press, **2026**; ISBN: 9780443239052
2. **Muchalski, H.**.; Johnston, J. N. Aziridination. In *Science of Synthesis: Stereoselective Synthesis*; de Vries, J. G., Ed.; Thieme: Stuttgart, **2011**; Vol. 1, pp 155–184

## Conference Presentations

**Contributed Oral Presentations**
(†undergraduate co-author, **presenting author**)

1. **Muchalski, H.**; *In with the old, In with the new: Lowering the barriers to incorporation of green chemistry into organic chemistry laboratorys*, 27th Annual Green Chemistry & Engineering Conference, Long Beach, CA, June 13-15, 2023.
2. **Datsko, J. J.**; Muchalski, H. *Competition Kinetics of Peroxidation with $^{19}$F NMR Spectroscopy: Synthesis and Characterization of Hydroperoxides of para-Fluoroallylbenzene*; 45th Annual Central California Research Symposium, Fresno, CA, 2024.
3. **Closser, K. D.**; Hawker, M. J.; Muchalski, H.; *Quantized Grading: An Ab Initio Approach to Using Specifications-Based Grading in Physical Chemistry*; 2022 Biennial Conference on Chemical Education, West Lafayette, IN, July 31–August 4, 2022.
4. **Dillon, C. C.**†; Keophimphone, B.†; Sanchez, M.†; Kaur, P.†; Muchalski, H. *Gold(I)-NHC-Catalyzed Synthesis of Benzo[b]thiophenes from 2-Alkynylthioanisoles*, 40th Annual Central California Research Symposium, Fresno, CA, May 1, 2019.
5. **Olvera, A. C.**†; Ramos Flores, J.†; Muchalski, H. *Towards Understanding of Peroxidation of Mammalian Sterols: Microwave-Assisted Synthesis of 7-Dehydrocholesterol Isomers*, 38th Annual Central California Research Symposium, Fresno, CA, April 18-19, 2017.
6. **Muchalski, H.**.; Lamberson, C. R.; Levonyak, A. J.; Xu, L.; Porter, N. A. *Does quantum mechanical tunneling make free radical peroxidation favorable?*, Abstracts of Papers, 248th ACS National Meeting, San Francisco, CA, August 10-14, 2014.
7. **Muchalski, H.**; Xu, L; Porter, N. A. *Kinetic isotope effect of deuterium-reinforced 7-dehydrocholesterol in tocopherol-mediated free radical chain oxidation*, Abstracts of Papers, 247th ACS National Meeting, Dallas, TX, March 16-20, 2014.

**Invited Talks**

1. ACS Campaign for a Sustainable Future Chemistry Education Summit: Reimagining Chemistry Education, Washington, DC, December 7–8, 2023. *Green Chemistry Module for Organic Chemistry: Carboxylic Acids and Derivatives*
2. California State University, Fresno, CA. *Sabbatical Report*, October 10, 2022.
3. San Jose State University, San Jose, CA. *Gold(I)-NHC-Catalyzed Synthesis of Benzofurans from 2-Alkynyl Aryl Ethers*, February 26, 2019.
4. University of Tulsa, Tulsa, OK. Jan 2015.
5. Kent State University, Kent, OH. *Kinetic Isotope Effect in Peroxidation of Lipids and Sterols*, Jan 2015.
6. California State University, Fresno, CA. *Kinetic Isotope Effect in Peroxidation of Lipids and Sterols*, Jan 2015.
7. Murray State University, Murray, KY. *Kinetic Isotope Effect in Peroxidation of Lipids and Sterols*, Nov 2014.
8. University of Tampa, Tampa, FL. *Kinetic Isotope Effect in Peroxidation of Lipids and Sterols*, Dec 2014.
9. University of Lodz, Lodz, Poland. *Stereospecific Reactions of α-Aminodiazonium Intermediates: Mechanistic Studies, New Reaction Discovery and Application to a Bidirectional Synthesis of (+)-Zwittermicin A*, May 2012.

10. Wroclaw University of Technology, Wroclaw, Poland. *Stereospecific Reactions of α-Aminodiazonium Intermediates: Mechanistic Studies, New Reaction Discovery and Application to a Bidirectional Synthesis of (+)-Zwittermicin A*, May 2012.

**Poster Presentations**

†undergraduate student; ‡graduate student

1. Breine, F. J.†; Muchalski, H.; *Synthesis and determination of the rate constant of inhibition of 2-dimethylamino-4,6-dimethyl-5-pyrimidinol using $^{19}$F NMR peroxyl radical clock*; 2026 Central California Research Symposium, Fresno, CA, April 22, 2026.

2. Sadeddin, M. J.†; Muchalski, H.; *Synthesis and characterization of peroxidation products originating from 3-(4-fluorophenyl)prop-2-enyl radical*, 2026 Central California Research Symposium, Fresno, CA, April 22, 2026.
3. Jang, H.†; Bolton, J.Z.;† Muchalski, H.; *Identification of amino acids using thin-layer chromatography augmented by $^{19}$F NMR spectroscopy*, 2026 Central California Research Symposium, Fresno, CA, April 22, 2026.
4. Breine, F. J.†; Muchalski, H.; *Synthesis and determination of the rate constant of inhibition of 2-dimethylamino-4,6-dimethyl-5-pyrimidinol using $^{19}$F NMR peroxyl radical clock*, 2026 Spring National Meeting of the American Chemical Society, Atlanta, GA, United States, March 22–26, 2026
https://doi.org/10.1021/scimeetings.5c11672
5. Sadeddin, M. J.†; Muchalski, H.; *Synthesis and characterization of peroxidation products originating from 3-(4-fluorophenyl)prop-2-enyl radical*, 2026 Spring National Meeting of the American Chemical Society, Atlanta, GA, United States, March 22–26, 2026;
https://doi.org/10.1021/scimeetings.5c11731
6. Cano, A. S.†; Muchalski, H.; *Evaluation of pyridyl thioesters as carbonyl activator in intramolecular Friedel–Crafts acylation reaction*, 2025 Western Regional Meeting of the American Chemical Society, San Jose, CA, October 26, 2025.
7. Breine, F. J.†; Muchalski, H.; *Synthesis and determination of the rate constant of inhibition of 2-dimethylamino-4,6-dimethyl-5-pyrimidinol using $^{19}$F NMR peroxyl radical clock*, 2026 Central California Research Symposium, Fresno, CA, April 22, 2026.
8. Sadeddin, M. J.†; Muchalski, H.; *Determination of H-atom transfer rate constants of phenolic antioxidants using $^{19}$F NMR peroxyl radical clock*, 46th Annual Central California Research Symposium, Fresno, CA, April 23, 2025.
9. Breine, F. J.†; Lim, C. C.†; Muchalski, H.; *Less solvent, more NMR skills: Development of green experiments for organic chemistry laboratory courses*, 46th Annual Central California Research Symposium, Fresno, CA, April 23, 2025.
10. Aguilar, J.‡; Cano, A. S.†; Muchalski, H.; *Synthesis and evaluation of 2-pyridyl thioester as reactive $^{19}$F-tags for identification and quantification of amino acids*, 46th Annual Central California Research Symposium, Fresno, CA, April 23, 2025.
11. Sadeddin, M. J.†; Muchalski, H.; *Determination of H-atom transfer rate constants of phenolic antioxidants using $^{19}$F NMR peroxyl radical clock*, Abstracts of Papers, ACS Spring Meeting 2025, San Diego, CA, United States, March 23–27 2025.
12. Breine, F. J.†; Lim, C. C.†; Muchalski, H.; *Less solvent, more NMR skills: Development*

*of green experiments for organic chemistry laboratory courses*, Abstracts of Papers, ACS Spring Meeting 2025, San Diego, CA, United States, March 23–27 2025.

13. Muchalski, H.; Medina Ramos, K. G.[†]; Datsko, J.J.[‡]; Bustos, K. N.; Abajian, A. G.[†]; *Competition kinetics of autoxidation with NMR spectroscopy: Development of* $^{19}$*F NMR peroxyl radical clock*, Abstracts of Papers, ACS Fall Meeting 2024, Denver, CO, United States, August 18–22 2024.
14. Aguilar-Cisneros, J.[†]; Garcia, I.C.[†]; *Cascade Reaction of Benzofurans via Sonogashira Cross-Coupling in a Micellar Medium*, 45th Annual Central California Research Symposium, Fresno, CA, April 24, 2024.
15. Medina Ramos, K. G.[†]; Abajian, A. G.[†]; *Efficient Synthesis of Fluorine-Tagged tert-Butyl Peroxyester Precursor of Allylbenzene Radical*, 45th Annual Central California Research Symposium, Fresno, CA, April 24, 2024.
16. Legget, A. A.[†]; *Composition of Clove Oil Isolated by Steam Distillation and Liquid Carbon Dioxide Extraction*, 45th Annual Central California Research Symposium, Fresno, CA, April 24, 2024.
17. Stevens, M. D.[‡]; Muchalski, H.; *Synthesis of Benzothiophenes in Water Catalyzed by Gold(I)–NHC Complexes*, 35th CSU Annual Biotechnology Symposium, Santa Clara, CA, January 13–14, 2023.
18. Pisor, J. W.[‡]; Mamo, K.[†]; Garcia, I.C.[†]; Muchalski, H.; *Synthesis of Benzofurans via Au(I)-Catalyzed Cyclization of 2-Alkynyl Phenol Derivatives*, 35th CSU Annual Biotechnology Symposium, Santa Clara, CA, January 13–14, 2023.
19. Pisor, J. W.[‡]; Garcia, I. C.[†]; Mamo, K.[†]; Muchalski, H.; *Synthesis Of 2-Substituted Benzofurans From 2-Alkynyl Aryl Ethers Catalyzed By Gold(I)–N-Heterocyclic Carbene Complexes*, Annual Biomedical Research Conference for Minority Students (ABRCMS), Long Beach, CA, November 9-12, 2022.
    **Award:** ABRCMS 2022 Steering Committee Presentation Award (Kirubel Mamo).
20. Pisor, J. W.[‡]; Garcia, I. C.[†]; Mamo, K.[†]; Muchalski, H.; *Synthesis Of 2-Substituted Benzofurans From 2-Alkynyl Aryl Ethers Catalyzed By Gold(I)–N-Heterocyclic Carbene Complexes*, Society for Advancement of Chicanos/Hispanics & Native Americans in Science (SACNAS), Puerto Rico, October 27-29, 2022.
21. Lichtenstein, D. A.[†]; Dillon, C. C.[‡]; Le, Q.[‡]; Muchalski, H.; *Gold(I)–NHC-catalyzed synthesis of benzofurans via migratory cyclization of 2-alkynylaryl benzyl ethers*, College or Science and Mathematics Virtual Research Showcase, May 8–15, 2020.
22. Lichtenstein, D. A.[†]; Dillon, C. C.[‡]; Le, Q.[‡]; Muchalski, H.; *Gold(I)–NHC-catalyzed synthesis of benzofurans via migratory cyclization of 2-alkynylaryl benzyl ethers*, Abstracts of Papers, 259th ACS National Meeting & Exposition, Philadelphia, PA, March 22-26, 2020, CHED-1230
23. Lichtenstein, D. A.[†]; Dillon, C. C.[‡]; Le, Q.[‡]; Muchalski, H.; *Gold(I)–NHC-catalyzed synthesis of benzofurans via migratory cyclization of 2-alkynylaryl benzyl ethers*, 32nd CSU Annual Biotechnology Symposium, Santa Clara, CA, January 16–18, 2020.
24. Phasakda, A.[†]; Muchalski, H. *Studies of directed gold(I)-catalyzed hydrocarboxylation of usymmetrical alkynes* 40th Annual Central California Research Symposium, Fresno, CA, May 1, 2019.
25. Lichtenstein, D. A.[†]; Le, Q.[‡ †]; Muchalski, H. *Development of gold(I)-catalyzed synthesis of benzofurans via gold(I)-catalyzed cyclization of 2-alkynyl ethers* 40th Annual

Central California Research Symposium, Fresno, CA, May 1, 2019.

26. Pisor, J.W.†; Avalos, D. †; Sanchez, M.†; Muchalski, H. *Development in the syntheses of isoquinolinones via gold(I)-catalyzed cyclization of 2-alkynyl Weinreb amides* 40th Annual Central California Research Symposium, Fresno, CA, May 1, 2019.
27. Waite, J.A.†; Bustos, K. †; Ewing, A. L.†; Muchalski, H. *Substrate scope studies of the gold(I)-catalyzed synthesis of 2,3-disubstituted benzofurans* 40th Annual Central California Research Symposium, Fresno, CA, May 1, 2019.
   **Award:** Outstanding Poster Presentation in Chemistry (San Joaquin Valley Local Section of ACS)
28. Dillon, C. C.†; Keophimphone, B.†; Sanchez, M.†; Kaur, P.†; Muchalski, H. *Synthesis of 2-substituted benzo[b]thiophenes via gold(I)–IPr hydroxide- catalyzed cyclization of 2-alkynyl thioanisoles*, Abstracts of Papers, 257th ACS National Meeting & Exposition, Orlando, FL, Mar. 31-Apr. 4, 2019 (2019), ORGN-0099
29. Keophimphone, B.†; Sanchez, M.†; Muchalski, H. *Scope of the Gold(I)-IPr-OH-Catalyzed Synthesis of Benzo[b]thiophenes*, 31nd CSU Annual Biotechnology Symposium, Orange County, CA, January 3–5, 2019.
30. Sanchez, M.†; Phasakda, A.†; Muchalski, H. *Synthesis of Benzo[b]thiophenes Catalyzed by Gold(I)-IPr-Cl Complex* 39th Annual Central California Research Symposium, Fresno, CA, April 25, 2018.
31. Kaur, P.†; Dillon, C. C.†; Muchalski, H. *Optimization of Gold-Catalyzed Cyclization of 2-Alkynylthioanisole to 2-Phenylbenzo[b]thiophene* 39th Annual Central California Research Symposium, Fresno, CA, April 25, 2018.
   **Award:** Outstanding Poster Presentation in Chemistry (College of Science and Mathematics)
32. Hedgpeth, H.†; Sanchez, M.†; Gomez, J.†; Muchalski, H.; Person, E.*Effective Treatment of Laboratory Mercury Waste Using Polymer Made From Sulfur and Canola Oil* 39th Annual Central California Research Symposium, Fresno, CA, April 25, 2018.
33. Le, Q.; Watters, R. R.†; Muchalski, H. *Synthesis of Solution Stable Sulfenic Acids*, 38th Annual Central California Research Symposium, Fresno, CA, April 18–19, 2017.
   **Award:** Outstanding Oral or Poster Presentation in Chemistry (San Joaquin Section of the ACS)
34. Olvera, A.C.†; Ramos Flores, J.†; Muchalski, H. *Towards Understanding of Peroxidation of Mammalian Sterols: Microwave-Assisted Synthesis of 7-Dehydrocholesterol Isomers*, Abstracts of Papers, 253rd ACS National Meeting, San Francisco, CA, April 2-6, 2017 (2017), ORGN-521
35. Olvera, A.C.†; Ramos Flores, J.†; Muchalski, H. *Microwave-Assisted Synthesis of 7-Dehydrocholesterol Isomers for Structure–Oxidizability Relationship Studies*, SACNAS 2016
36. Muchalski, H. *Kinetic Isotope Effect of Deuterium-Reinforced 7-Dehydrocholesterol in Toco-pherol-Mediated Free Radical Chain Oxidation*, Vanderbilt Institute of Chemical Biology Symposium, 2013
37. Muchalski, H. *Stereospecific Reactions of α-Amino-β-Diazonium Intermediates: Mechanistic Studies, New Reaction Discovery and Application to a Bidirectional Synthesis of (+)-Zwittermicin A*, Gordon Research Conferences: Organic Reactions & Processes, 2011

*See my personal webpage for a complete list of conference presentations*

## Grants

NSF Chemical Mechanism, Function, and Properties (CMFP) 2025
Facilitating Research at Predominantly Undergraduate Institutions (RUI)
*RUI: Competition Kinetics of Peroxidation with Quantitative NMR Spectroscopy*
Under review; requested budget: $542,409, 3 years

National Science Foundation Chemical Catalysis Program 2023
Research Opportunity Award (ROA-PUI) Supplement to CHE 2152566
*New Advances in Sustainable Catalysis*
Visiting Associate Researcher at UC Santa Barbara with Prof. Bruce H. Lipshutz
**Awarded $42,354**

ACS Petroleum Research Fund 2022–2026
Undergraduate Research Grant (Physical Organic Chemistry)
*Development of Heteronuclear Quantitative NMR Assay for Direct Peroxyl Radical Clock Kinetics*
**Awarded $70,000**

Organic Syntheses PUI Grant
*Metal-Catalyzed Preparation of tert-Butyl Peroxyesters of 4-Aryl-3-Butenoic Acids*
Declined; requested budget: $8,000 2022

American Chemical Society Division of Organic Chemistry 2022
Summer Undergraduate Research Fellowship
*Metal-Catalyzed Preparation Fluorine-Tagged Peroxyesters of 4-Aryl-3-Butenoic Acids*
**Awarded $6,000** (to Karen Medina Ramos)

National Science Foundation CHE/CSDM-B 2022
*RUI: Development of Peroxyl Radical Clock Methodology Using Quantitative Heteronuclear NMR*
Declined; requested budget: $441,626, 3 years

CSUBiotech‡ Faculty–Graduate Student Research Collaboration 2022
*Efficient Synthesis of Benzofuran Heterocycles Catalyzed by Gold(I)-NHC Complexes*
**Awarded $10,000**

CSM Course Based Undergraduate Research Experience (CURE) 2022
*Green Chemistry and Catalysis: Synthesis of Heterocycles in Water*
**Awarded $4,000**

CSUPERB‡ Graduate Student Research Restart Program 2021
*Synthesis NHC-Gold Complexes for Synthesis of Heterocycles in Water*
Declined; requested budget: $6,500

CSUPERB‡ COVID-19 Research Recovery Grant 2021
*New Methodologies for Free Radical Oxidation Kinetics and Synthesis of Silyl Enol Ethers*
**Awarded $1,067**

CSUPERB‡ New Investigator Grant 2018–2020
*Metal-Catalyzed Synthesis of Enol Esters for Controlled Release of Pheromonones*
**Awarded $15,000**

Dreyfus Teacher Scholar Award 2020
*Gold-Catalyzed Synthesis of Heterocycles*
Declined; requested budget: $75,000

CSUPERB‡ Presidents' Commission Scholar Program 2018
*Synthesis and Evaluation of the Scope of Cyclization of 2-Alkynylthioanisoles to Benzo-[B]Thiophenes Catalyzed by Gold(I)-N-Heterocyclic Carbene Complexes*
**Awarded $8,000** (to Bagieng Keophimphone) 2018

National Science Foundation CHE/CSDM-B 2017
*RUI: Organogold Chemistry Involving Siloxides and Silanols*
Declined; requested budget: $198,969 over 2 years

CSUPERB‡ New Investigator: 2016
*Development of Gold-Catalyzed Synthesis of Z-Vinyl Acetates*
Declined, requested budget: $15,000 2017

National Science Foundation 2016
*RUI: Synthesis and Characterization of Stable Sulfenic Acids*
Declined, requested budget: $337,624 over 3 years

CSUPERB‡ New Investigator 2016
*New Strategies for the Synthesis of Deuterium-Reinforced Fatty Acids*
Declined, requested budget $15,000

American Chemical Society Petroleum Research Fund 2016
Undergraduate New Investigator Grant
*Synthesis of Sulfenic Acid-Based Antioxidants*
Declined, requested budget: $70,000 over 3 years

‡CSUBiotech (formerly CSUPERB) - California State University Program for Education and Research in Biotechnology

## Teaching Experience

**Graduate Courses**

*Advanced Research Techniques (CHEM 260)*
Sp20

*Strategies and Tactics in Organic Synthesis (CHEM 240T)*
Fa19, Fa22, Fa25, Fa26

*Seminar in Chemistry (CHEM 280)*
Fa18

*Topics in Advanced Organic Chemistry (CHEM 240T)*
Fa15

**Undergraduate Courses** (H = Honors Course; † = Virtual)

*Organic Chemistry 1 (CHEM 128A)*
Fa15, Fa16, Fa17, Fa18, Su19, Fa19, Fa20, Fa21, Sp23 (x2), Fa23, Fa24

*Organic Chemistry 2 (CHEM 128B)*
Sp16, Sp17, Sp19, Fa20†, Su21†, Su22

*Organic Chemistry Laboratory 1 (CHEM 129A)*
Sp16, Fa19 (x2), Fa20† (x2), Fa21 (2x), Fa26; Coordinator: 2022–present

*Organic Chemistry Laboratory 2 (CHEM 129B)*
Fa16, Fa17, Sp18 (x2), Su20†, Sp21†, Fa22, Sp23, Fa23, Sp24, Fa24, Sp25, Sp26; Coordinator 2020–present

*Introduction to Biomedical Research (CHEM 140T)*
Fa23

*Research Techniques (CHEM 160H)*
Sp20

*Seminar in Chemistry (CHEM 180H)*
Fa18

## Advising

**Graduate students**
*Thesis Chair (9)*: Quang Le, Christopher C. Dillon, Karina N. Bustos (NSF B2D 2020), Jeremy W. Pisor, Michael D. Stevens, Kiersten Friesen, Jason Datsko, Jorge Aguilar-Cisneros, Eric Lockton.

*Thesis Committee Member (14)*: Denis Ashong, Jasmine Hang, Roxana Gonzalez Dorado, Stan Gomez, Esveidy Osegura Nava, Dennis Ashong, Inderpal Sekhon, Christian Dwyer, Eric Munoz, Sitong Wu, Ziran Jiang, Pravien Rajaram, Keeton Montgomery, Mericarmen Gonzalez, Anthony Hinde, Xiang Li

**Undergraduate students**
*Honors Thesis Advisor (11)*: Felix Breine, HeeWon (Ann) Jang, Mariah Sadeddin, Ani Abajian, Isabella Garcia, Montaser Ahmad, Elizabeth Herren, Simrit Dhindsa, Alexander Ewing, Bagieng Keophimphone, Parveen Kaur

*NIH RISE Program Advisor (5)*: Mariah Sadeddin, Angelia Leggett, Isabella Garcia, Kirubel Mamo, Angel Rojas.

*Fresno City College Chemistry Certificate Internship Students (4)*: HeeWon (Ann) Jang, Felix Breine, Alan Lopez, Angel Rojas.

*ACS Project SEED High School Students (3)*: Jonathan Jimenez (2017), Aliyah Lerma (2018), Jasmine Ortigoza (2022).

## Professional Development

| | |
|---|---|
| **Grading Conference** | June 11–13, 2025 |
| **Markers of Excellence in ACS Approved Programs** | March 17, 2024 |
| **Grading Conference** | June 13–15, 2024 |
| **General Chemistry Curriculum Reform Workshop** | March 26, 2023 |
| **Grading Conference** | June 13–15, 2023 |
| **CPARS Reviewer Training for new CPT Members** | 2023 |
| **ACS Green & Sustainable Chemistry Module Development** | 2021–2025PRESENT |
| **22nd Annual R. Bryan Miller Symposium** | March 9–10, 2022 |
| **Mastery Grading Conference (Higher Ed STEM Focus)** | June 3–4, 2022 |
| **HyFlex Course Institute (facilitator)** | Summer 2021 |

| | |
|---|---|
| **Mastery Grading Conference (Higher Ed STEM Focus)** | **June 3–4, 2022** |
| June 11–12, 2021 | |
| **HyFlex Course Institute (participant)** | Spring 2021 |
| **Advanced Quality Learning and Teaching (QLT)** | Summer 2020 |
| **Introduction to Teaching Online Using (QLT)** | Spring 2020 |
| **Mastery Grading Conference** | June 5–6, 2020 |
| **Transforming STEM Teaching Faculty Learning Program** | |
| UC/CSU program supported by the NSF (DUE #1626624) | 2018 |
| **New Faculty Workshop** | |
| ACS–Cottrell Scholars Collaborative, Washington, DC, | August 3-5, 2017 |
| **Active Learning in Organic Chemistry** | |
| NSF cCWCS Mini-workshop, Atlanta, GA, | June 12–15, 2017 |
| **Early Career Investigator Workshop** | |
| NSF Division of Chemistry, Arlington, VA, | March 20-21, 2017 |
| **Certificate in College Teaching** | |
| Center for Teaching, Vanderbilt University | 2014 |

## Leadership and Service

**Fresno State**

| | |
|---|---|
| University Graduate Curriculum Subcommittee (Chair) | 2022–2025 |
| University Graduate Curriculum Subcommittee (member) | 2019–2025 |
| University Campus Planning Committee | 2023–2026 |
| Department Representative to the Academic Senate | 2018–2021 |
| Advisor to the ACS Student Chapter | 2017–2021 |
| College Curriculum Subcommittee | 2017–2019 |
| Department Safety Committee (member) | 2016–present |
| Department Safety Committee (chair) | 2025–present |
| Department Graduate Committee (member) | 2016–present |

**American Chemical Society**

| | |
|---|---|
| Committee on Professional Training | 2023–present |
| Chair, San Joaquin Valley Local Section | 2025 |
| Chair-Elect, San Joaquin Valley Local Section | 2024 |
| Chair, District VI Councilor Caucus | 2023 |
| Chair-Elect, District VI Councilor Caucus | 2022 |
| Councilor, San Joaquin Valley Local Section | 2018–2023 |
| National Chemistry Week Outreach Coordinator | 2018–2024 |
| Chemists Celebrate Earth Week Outreach Coordinator | 2018–2024 |
| Treasurer, San Joaquin Valley Local Section | 2016–2017 |

**Green Chemistry Institute**

| | |
|---|---|
| Nina McClelland Memorial Award Committee | 2024–present |
| Joseph Breen Memorial Fellowship Committee | 2024–present |
| Kenneth G. Hancock Memorial Award Committee | 2024 |

| | |
|---|---|
| Career Achievement in Green Chemistry Education Award Committee | 2025 |
| Heh-Won Chang PhD Fellowship Committee | 2025 |
| Green Chemistry School Fellowship Committee | 2025 |

**Peer Reviewer**

European Journal of Inorganic Chemistry
Chemistry - An Asian Journal
Organic and Biomolecular Chemistry
RSC Advances

## Affiliations

**American Chemical Society**
*Member* — 2012-PRESENT
**Vanderbilt University**, Nashville, TN
*Visiting Scholar* — 2015–2021
**University of California Santa Barbara**
*Visiting Associate Researcher* — 2023